

Harrison Doan

Global Military Expenditure – Intelligence Report

Dataset: SIPRI-Milex-Data (1949-2025)

Produced: May 21st, 2026

Executive Summary

This project analyzed historical global military expenditure trends using the SIPRI Military Expenditure Database to identify regions and subregions experiencing significant acceleration in defence spending, burden shifts, and long-term growth. Using SQL, Excel, and Power BI, the analysis identified substantial post-2020 increases in military expenditure growth across Eastern Europe, Central and Western Europe, and South East Asia, alongside materially improved NATO burden-sharing compliance following 2022. The findings suggest a broader global shift toward military modernization, alliance strengthening, and strategic rearmament in response to evolving geopolitical instability and changing security environments.

Objective

The objective of this project is to analyze historical global military expenditure trends and identify regions, subregions, and countries where spending has accelerated, governments have taken greater military spending burdens relative to economic output, and uncharacteristic military expenditure has emerged. Using SQL, Power BI, and Excel, the project aims to generate intelligence-style analytical insights to support strategic risk assessment, geopolitical monitoring, and evaluation of future defence policy.

Data Description

The analysis utilized the Stockholm International Peace Research Institute (SIPRI) Military Expenditure Database, covering annual military expenditure data from 1949 to 2024. The dataset includes country and regional-level military spending metrics reported in current U.S. dollars (USD), military expenditure as a percentage of GDP, military expenditure as a percentage of government spending, and per-capita military expenditure.

Methodology

The analysis was conducted using Excel, SQL, and Power BI. Initial data cleaning and transformation were completed in Excel using Power Query, where the raw SIPRI military expenditure dataset was standardized and separated into multiple structured CSV files for database integration. Supporting Excel pivot tables, heatmaps, and charts were also created throughout the analysis process to assist with exploratory analysis and query validation.

Following the transformation stage, a relational SQL Server database was designed and implemented to support cross-relational analysis. Multiple parent and connecting tables were created using primary and foreign key relationships to enable efficient querying across countries, regions, subregions and expenditure metrics. Analytical analysis was then conducted in SQL using a combination of common table expressions (CTEs), window functions, conditional aggregation, CASE statements, and year-over-year growth calculations. The analysis focused on identifying expenditure acceleration trends, defence burden shifts, NATO burden-sharing patterns, and sustained military expenditure growth.

Finally, the SQL outputs were imported into Power BI, where additional data shaping was completed using the Power Query editor. A dynamic dashboard was developed to communicate key analytical findings through heatmaps, KPI indicators, trend analysis, and comparative regional visualizations.

Key Analytical Findings

Finding #1: Regional/Subregional acceleration intensified following 2020

As demonstrated in the Power BI subregional military expenditure acceleration heatmap, multiple regions experienced significant increases in military spending growth rates beginning in 2021. Eastern Europe, Central and Western Europe, and South East Asia displayed sustained year-over-year expenditure acceleration, while subregions such as Sub-Saharan Africa and Central Asia experienced periods of contraction and comparatively slower defence spending growth.

Eastern Europe experienced the most substantial acceleration during the period analyzed, with military expenditure growth sharply increasing in 2022 following the Russian invasion of Ukraine in February of the same year. The sustained acceleration observed across both Eastern Europe and Central and Western Europe suggests a broader regional response to rising geopolitical instability, heightened security concerns, and increased defence modernization efforts across Europe.

Finding #2: NATO burden-sharing improved materially after 2022

As demonstrated in the Power BI NATO compliance analysis, military expenditure burden-sharing among NATO member states increased significantly following 2022. The NATO compliance rate nearly doubled during that period, indicating a broader commitment among member nations to increase defence spending and strengthen collective security preparedness. The findings suggest that NATO members increasingly prioritized defence investment in response to the deteriorating European security environment and heightened geopolitical tensions following the Russian invasion of Ukraine.

In terms of alliance posture, the increase in NATO compliance rates suggests a broader distribution of defence responsibilities across member states rather than reliance on a limited number of major military powers. The trend also indicates a heightened perception of regional security risk among NATO members following 2022, particularly among states geographically closer to Eastern Europe and the Russian sphere of influence. Collectively, the findings suggest that the alliance has shifted toward a more unified and sustained defence posture in response to changing geopolitical conditions.

Finding #3: Unusually persistent long-term growth from smaller subregions

As depicted in the Power BI subregional acceleration and persistence bubble chart, several subregions exhibited both sustained and high-magnitude military expenditure growth

throughout the analyzed period. The visualization evaluated subregional spending trends using three primary dimensions: the total magnitude of 10-year expenditure growth, the number of years with positive year-over-year growth, and relative 2024 military expenditure size. Collectively, these indicators helped distinguish temporary expenditure spikes from more persistent long-term defence spending acceleration.

Subregions' positions in the upper right portion of the visualization demonstrated the strongest combination of sustained growth and long-term expenditure acceleration, indicating that military spending increases were not isolated to single-year geopolitical events. Instead, the findings suggest the presence of broader structural shifts in regional defence priorities and long-term strategic rearmament efforts. Larger bubble sizes additionally highlighted subregions where sustained acceleration was accompanied by significant overall expenditure levels, increasing their potential geopolitical and strategic significance.

Strategic Implications

The findings of the analysis suggest a broader global shift toward military modernization and strategic rearmament across several regions and subregions. Sustained increases in military expenditure acceleration, defence burden growth, and NATO burden-sharing compliance collectively indicate that many states are increasingly prioritizing defence preparedness in response to evolving geopolitical instability and heightened regional security concerns.

From a Canadian perspective, the findings reinforce the strategic importance of long-term defence investment, alliance participation, and military modernization planning. As global security dynamics continue to evolve, maintaining credible defence capabilities and strengthening interoperability with allied nations may become increasingly important for both national security and broader geopolitical positioning.

Additionally, rising regional instability and shifting alliance structures may create opportunities for Canada to strengthen strategic partnerships, defence cooperation initiatives, and economic relationships with allied and emerging partner nations. The sustained acceleration of military expenditure across multiple regions also suggests that defence procurement, intelligence analysis, cybersecurity, and strategic resource competition will likely remain increasingly important areas of policy focus in the coming years.

Finally, the analysis demonstrates how military expenditure data can serve as a useful proxy indicator for broader geopolitical tension, regional instability, and changing strategic priorities. Monitoring long-term expenditure acceleration patterns may therefore provide valuable insight into future security developments and emerging areas of geopolitical risk.

Limitations

Several limitations should be considered when interpreting the findings of this analysis. Military expenditure data does not directly measure military capability, operational readiness, or

strategic effectiveness, as spending efficiency and procurement priorities differ significantly across countries and regions. Additionally, the analysis primarily focused on expenditure acceleration and burden trends rather than broader geopolitical, economic, or diplomatic variables that may also influence defence spending behavior. Finally, while the SIPRI database is widely recognized and reliable, variations in national reporting standards and data availability may introduce minor inconsistencies across historical observations.

Conclusion

The analysis identified several significant shifts in global military expenditure patterns, particularly following 2020. Multiple regions and subregions demonstrated sustained expenditure acceleration, increased defence burden levels, and broader commitments toward military modernization and alliance preparedness. NATO burden-sharing compliance also improved materially following 2022, suggesting a more unified collective defence posture among member states.

Overall, the project demonstrates how military expenditure data can be leveraged through SQL, Excel, and Power BI to support intelligence-style analysis, identify emerging geopolitical trends, and monitor long-term changes in global defence priorities and strategic risk environments.